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Open Source Is Hot In The IT World

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maintain QoS and performance.

Ceph as a single storage cluster allows multiple data storage interfaces, which can utilise different hardware for different storage needs, managed by a single SDS system.

Ceph divides the OSDs into placement groups for the CRUSH algorithm. These placement groups can be combined together to form a pool, which is like a logical partition for storing the objects in Ceph. Pools can help differentiate between the storage hardware based on performance. Ceph also has cache-tiering, which helps in creating a pool of faster storage devices as cache storage for expensive read/write operations. This helps in improved performance and efficient utilisation of the storage hardware.

With OpenStack as the cloud platform, Ceph can be used as a Swift object store and Cinder block store utilising the same storage hardware for multiple needs. Ceph can be used with other cloud platforms like CloudStack, Eucalyptus and OpenNebula.

Ceph as a backup solution

We have been using RAID systems for redundancy and data backup purposes. With Ceph as the storage platform, data redundancy and recovery is managed using the RADOS storage system. RADOS has features like self-healing and self-managing, which help in recovering the data from lost OSD nodes using instant replication and rebalancing. Ceph also allows taking an incremental snapshot of block storage using RBD.

OpenStack has a Ceph backup driver, which is an intelligent solution of VM volumes. It backs up the volumes to Ceph's backend storage, and regularly performs an incremental backup on the volumes to maintain consistency.

CloudBerry Backup for Ceph is a popular tool, which helps in versatile control over Ceph's backup and recovery mechanism.

According to several research organisations like Gartner Inc., Ceph has made a strategic entry into the enterprise IT space and will prove to be the next big evolution in storage technology. With the current adoption rate, Ceph will soon surpass the existing storage solutions at enterprises. There is a lot of development expected to happen with Ceph, which will bring about significant performance improvements to match the current proprietary solutions. Even if you don't take future enhancements into consideration, Ceph is a storage platform that definitely needs to be looked at for your next big deployment. **END** 

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